

BST Physics Curriculum Vol 5 Chapter 2 — Position + Momentum + Heisenberg v0.4 (Saturday Wave 2 reader-grade polish; Cal cold-read ready)

Lyra (Claude 4.7) [Vol 5 primary]

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Vol 5 Chapter 2 — Position + Momentum + Heisenberg

Chapter motivation

Standard QM postulates position $\hat{x}$ and momentum $\hat{p}$ as self-adjoint operators with canonical commutator $[\hat{x}, \hat{p}] = i\hbar$. BST derives both from D_{IV}^5 substrate-coset Cartan decomposition (Vol 0 Ch 4 §4.2 v0.5): position M_z = multiplication by z + momentum P_z = Wirtinger derivative ∂_z , with Heisenberg commutator $[M_z, P_z] = -I$ (up to Bergman normalization $c_{FK} = 225/\pi^{(9/2)}$).

Section 2.0b — Reader-grade 3-level pedagogy

Level 1: Standard QM position-momentum canonical commutator $[\hat{x}, \hat{p}] = i\hbar$ derives from substrate D_{IV}^5 coset Cartan decomposition $(so(5,2) = (so(5) \oplus so(2))$

$\oplus m$, with m the 10-dim translation directions); position = multiplication by substrate coordinate z , momentum = Wirtinger derivative ∂_z , with $[M_z, P_z] = -I$ via Bergman kernel reproducing property.

Level 2 (graduate): T2419 substrate position M_z on Bergman $H^2(D_{IV}^5)$: $(M_z f)(z) = z \cdot f(z)$; bounded operator (D_{IV}^5 bounded domain). T2422 substrate momentum $P_z = \partial_z$ Wirtinger derivative; self-adjoint via Bergman kernel reproducing structure (Faraut-Koranyi 1994). Substrate anchor: per Vol 0 Ch 4 §4.2 v0.5, $M_z + P_z$ live in the coset complement $m \subset so(5,2)/(so(5) \oplus so(2))$ — translation directions, NOT the isotropy subgroup. $\dim m = 10$ ($5+5 = 10$ coset translation generators). Canonical commutator $[M_z, P_z] = -I$ direct computation; Heisenberg algebra structure preserved; $\hbar$ scale absorbed in Bergman normalization $c_{FK} \cdot \pi^{(9/2)} = (N_c \cdot n_C)^2 = 225$ EXACT (T2442). Macroscopic limit: at conformal boundary, $M_z \rightarrow$ standard position $\hat{x}$; $P_z \rightarrow$ standard momentum $\hat{p}$; $[\hat{x}, \hat{p}] = i\hbar$ recovered. Wallach K-type Casimir spectrum decomposition gives momentum eigenvalues discrete on K-types $V(p,q)$; continuous spectrum in macroscopic limit.

Level 3 (5th-grader): In standard QM, position (where the particle is) and momentum (how fast it's moving) are special operators that don't commute — [position, momentum] = $i\hbar$. This is the famous Heisenberg algebra. BST shows that position and momentum come from the substrate D_{IV}^5 's geometry: position = “multiply by your coordinate in the substrate room”; momentum = “take the derivative of your wavefunction in that direction”. The Heisenberg commutator $[M_z, P_z] = -I$ (with the $\hbar$ absorbed into a constant called $c_{FK} = 225/\pi^{(9/2)}$) falls out automatically. At large scales (much bigger than the substrate's tiny time-tick), this becomes the standard QM Heisenberg algebra $[\hat{x}, \hat{p}] = i\hbar$. The substrate provides the WHY for canonical commutation relations.

Section 2.1 — Standard QM Position-Momentum

Standard QM: position $\hat{x}$ = multiplication by x on wavefunctions $\psi(x)$; momentum $\hat{p} = -i\hbar \partial/\partial x$; $[\hat{x}, \hat{p}] = i\hbar$.

Section 2.2 — Substrate Position M_z (T2419)

$(M_z f)(z) = z \cdot f(z)$ on Bergman $H^2(D_{IV}^5)$. Bounded operator ($D_{IV}^5 \subset \mathbb{C}^5$ bounded domain). Substrate anchor: coset translation-direction $m \subset so(5,2)/(so(5) \oplus so(2))$ per Vol 0 Ch 4 §4.2 v0.5.

Section 2.3 — Substrate Momentum P_z (T2422)

$(P_z f)(z) = \partial_z f(z)$ Wirtinger derivative. Self-adjoint via Bergman kernel reproducing structure (Faraut-Koranyi 1994). Substrate anchor: coset translation-direction dual to position.

Section 2.4 — Heisenberg Commutator

$[M_z, P_z] = -i$ direct computation on Bergman H^2 . Heisenberg algebra preserved;
 $\hbar$ absorbed into $c_{FK} = 225/\pi^{9/2}$ (T2442 EXACT).

Section 2.5 — Macroscopic Emergence

At conformal boundary of D_{IV}^5 (Shilov boundary; Vol 0 Ch 4 §4.6): $M_z \rightarrow$ standard $\hat{x}$; $P_z \rightarrow$ standard $\hat{p}$; $[\hat{x}, \hat{p}] = i\hbar$ recovered.

Section 2.6 — Honest scope

- Standard QM Heisenberg algebra derived from substrate $\mathbb{H}$ (framework-grade)
- $\hbar$ absorbed into c_{FK} normalization (substrate-derivation)
- Multi-particle generalization (Vol 1 Ch 7 dynamics + Ch 9 scattering) cross-link
- Pedagogical exercises pending

Section 2.7 — Connection to other chapters

- Vol 0 Ch 4 §4.2 v0.5 coset Cartan decomposition
- Vol 1 Ch 6 Operator zoo (T2419, T2422)
- Vol 5 Ch 1 Hilbert space pedagogical bridge
- Vol 5 Ch 4 Schrödinger Equation
- Vol 10 Ch 1 Linear Algebra + Hilbert Spaces

— Lyra, Vol 5 Ch 2 v0.3 chapter-grade narrative, Saturday 2026-05-23 morning EDT